

E9 205 Machine Learning for Signal Processing

Dimensionality Reduction - I

21-08-2019

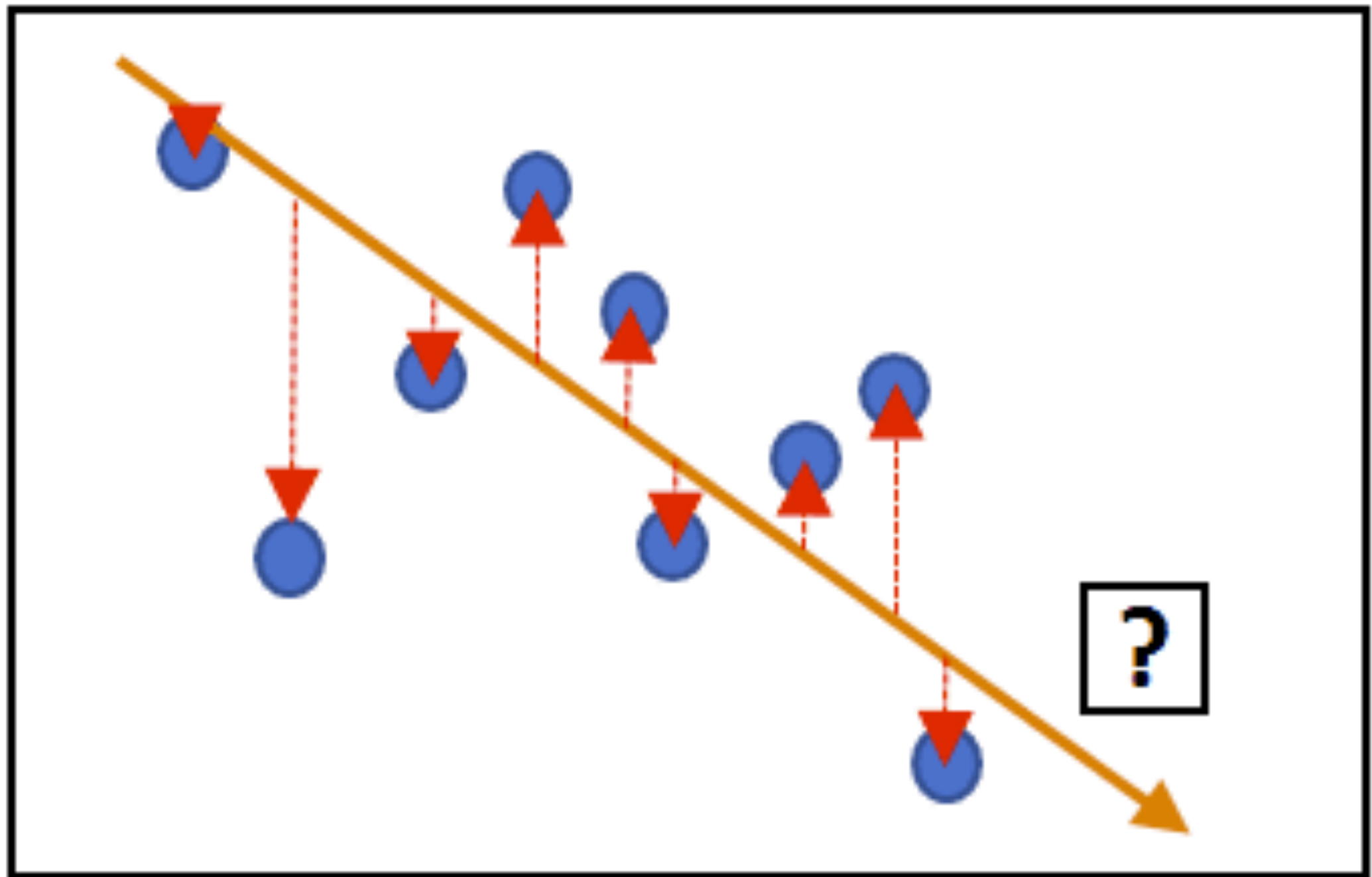
Instructor - Sriram Ganapathy (sriramg@iisc.ac.in)



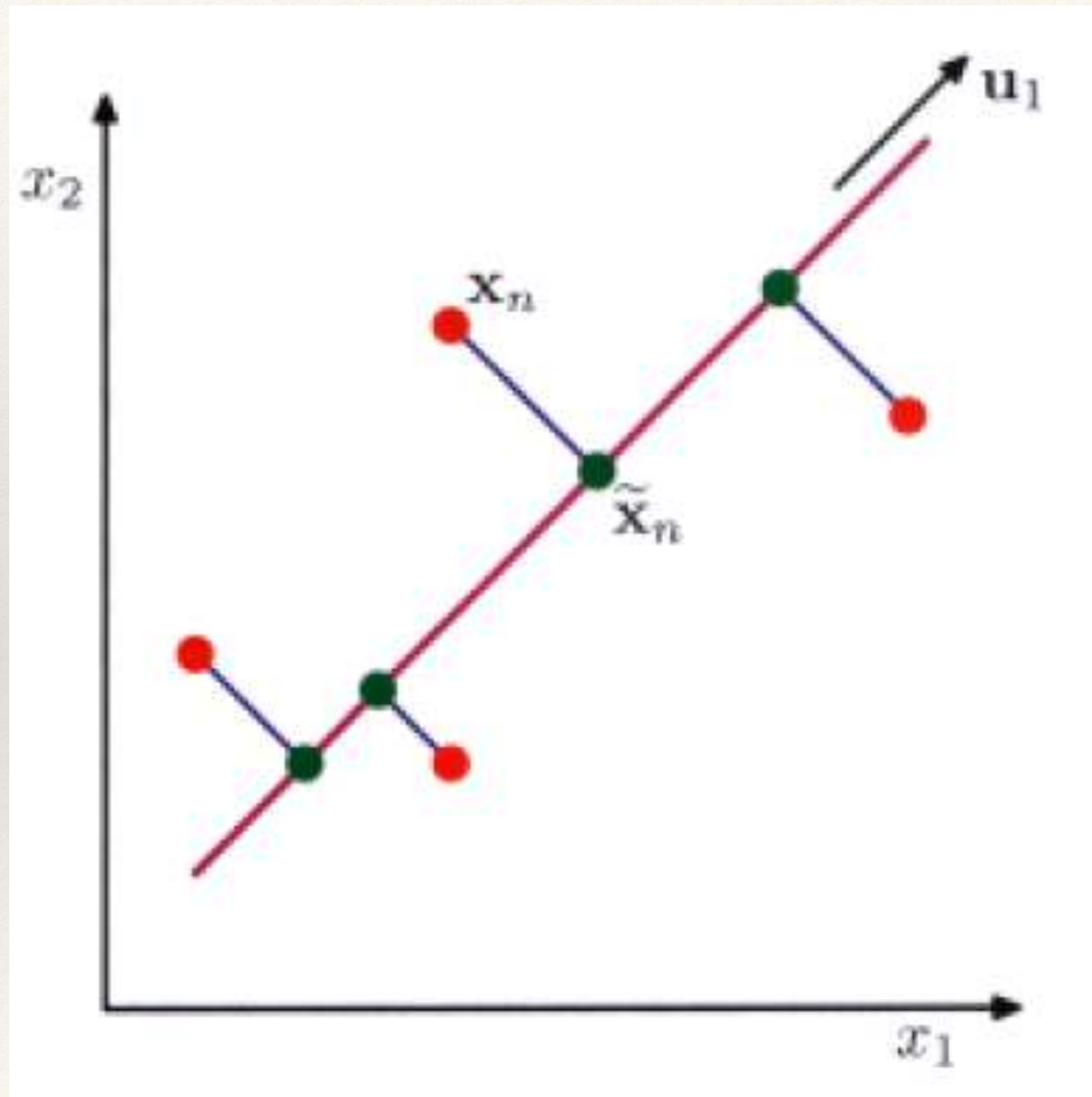
Principal Component Analysis

- ❖ Reducing the data $\mathbf{x}_n$ of dimension D to lower dimension $M < D$
- ❖ Projecting the data into subspace which preserves maximum data variance
 - ❖ Maximize variance in projected space
- ❖ Equivalent formulated as minimizing the error between the original and projected data points.

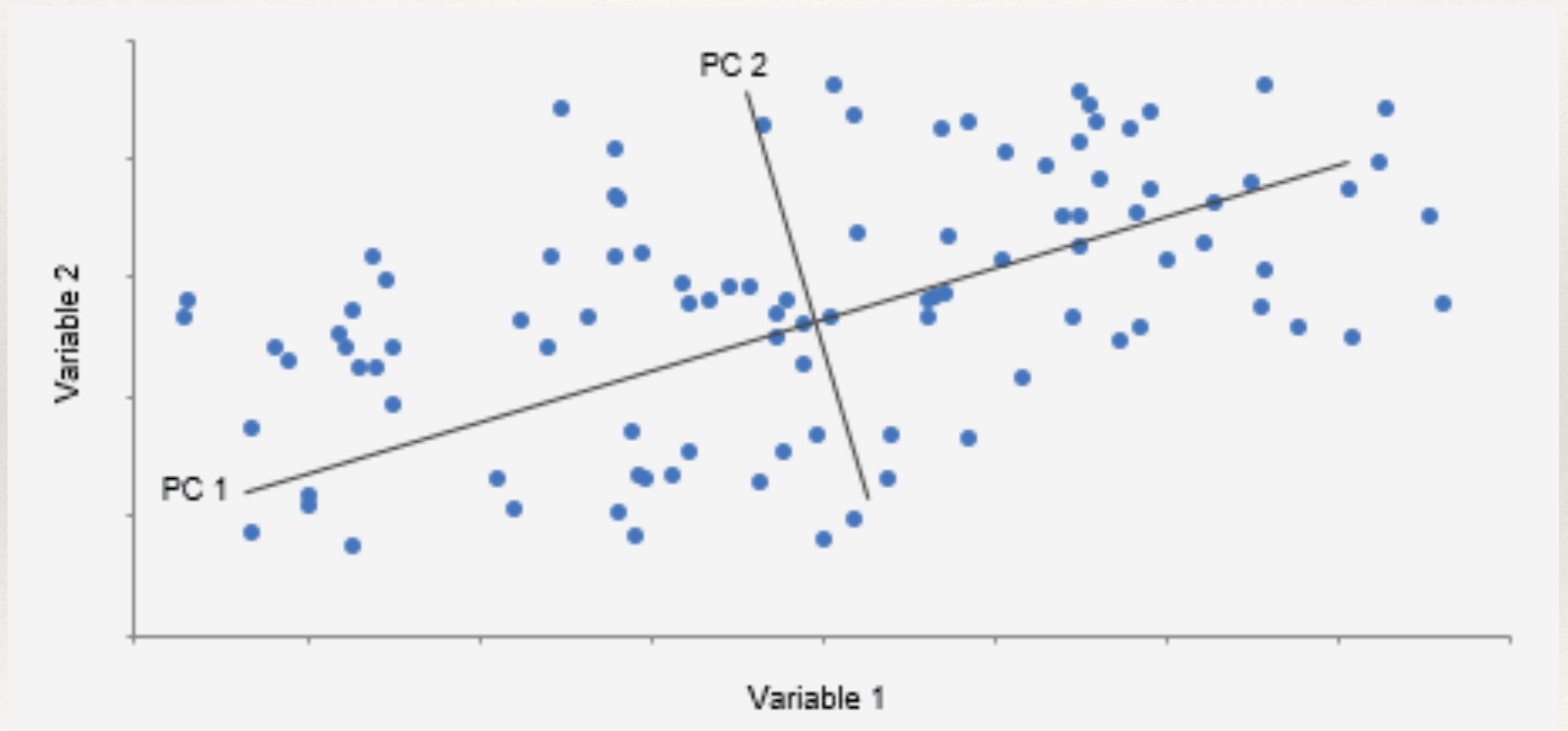
Direction of Maximum Variance



Minimum Error Formulation



PCA Example



Principal Component Analysis

- ❖ First M eigenvectors of data covariance matrix

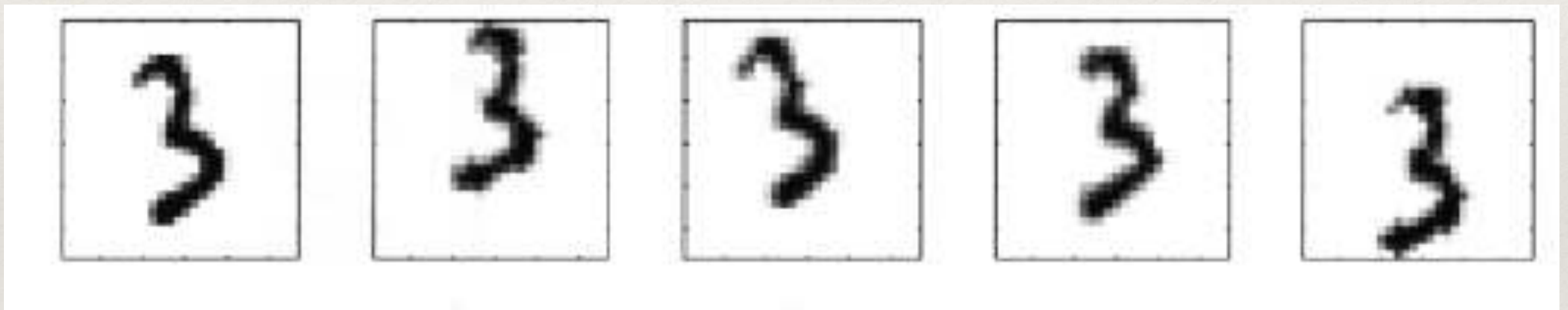
$$S = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T$$

- ❖ Residual error from PCA

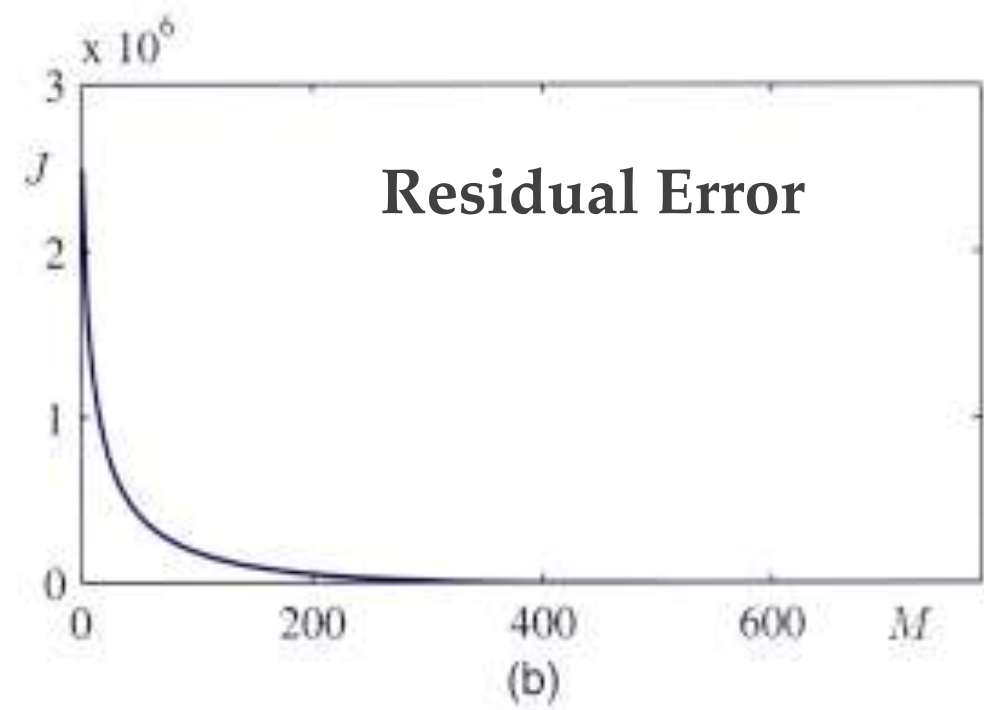
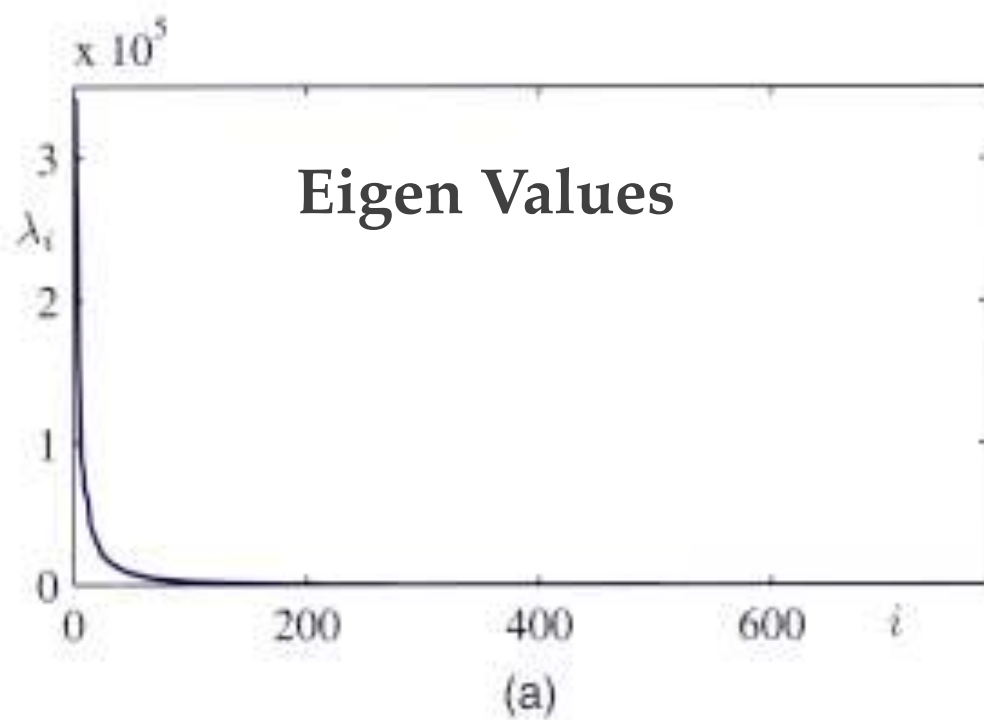
$$J = \sum_{i=M+1}^D \lambda_i$$

PCA

Handwritten digits used for PCA training...

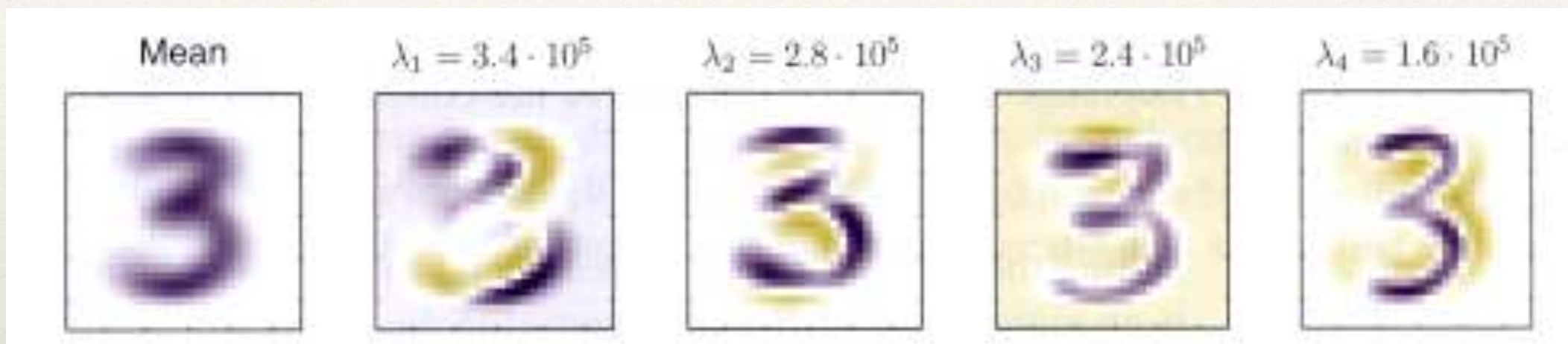


PCA

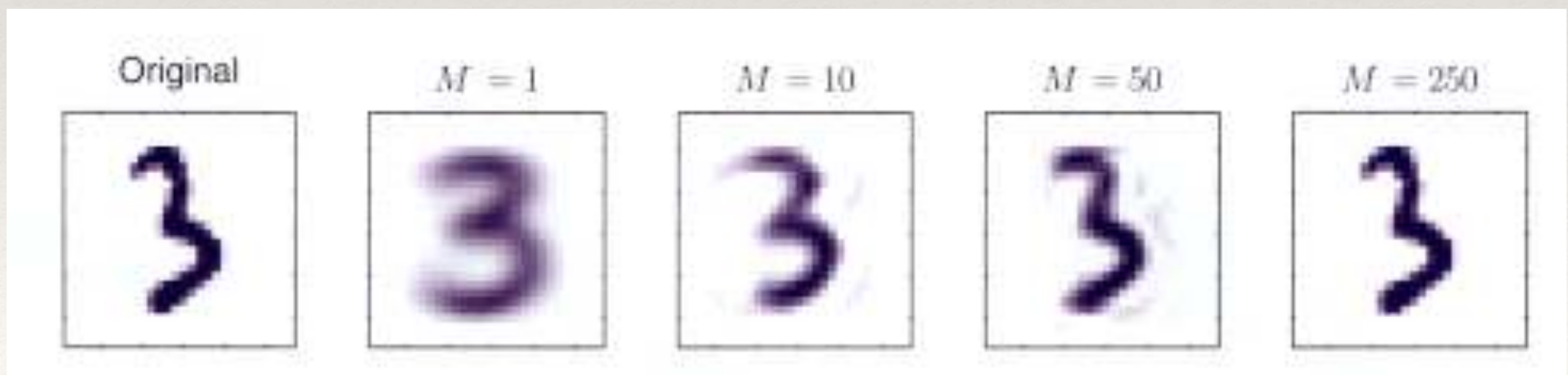


PCA - Reconstruction

Eigenvectors

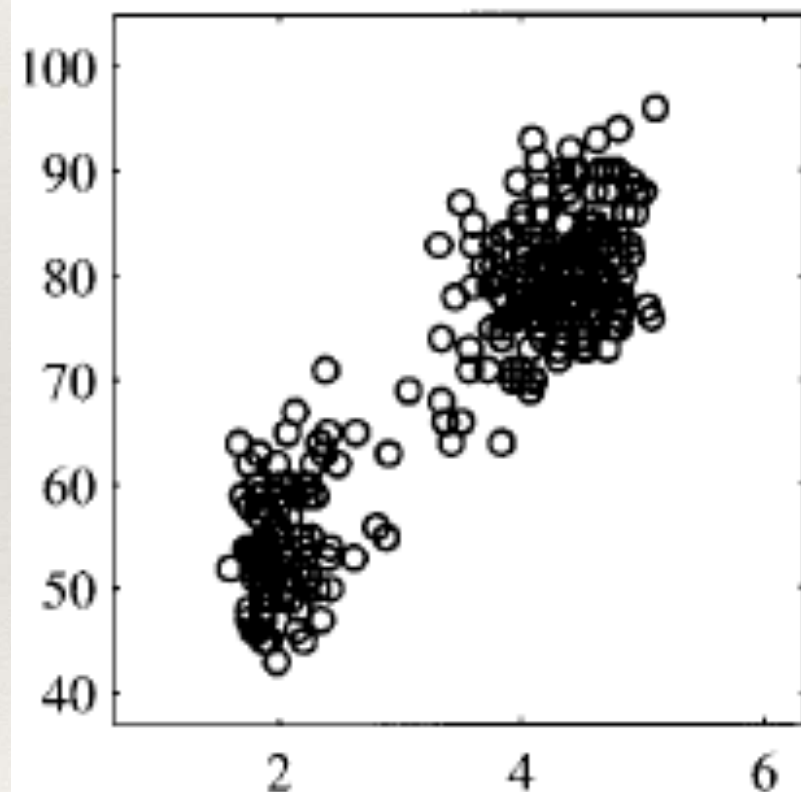


PCA - Reconstruction

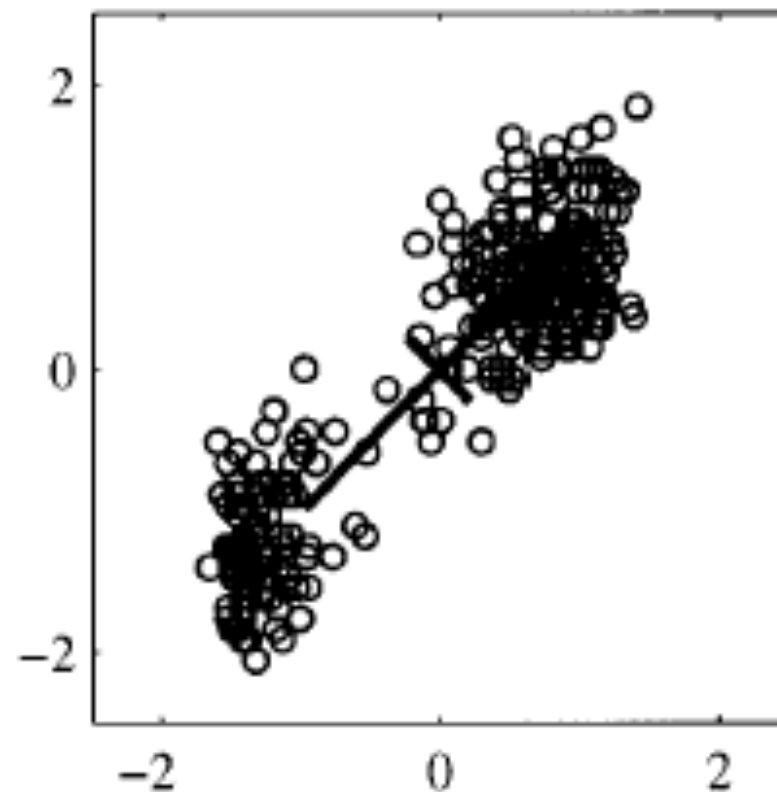


Whitening the Data

Original Data



Standardization



Whitening

